CTSpinoPelvic1K: A Unified CT Benchmark for Lumbosacral Transitional Vertebra Segmentation

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Abstract

Machine learning models for spine segmentation face two compounding blind spots: a *modality gap* (predominance of MRI-derived benchmarks despite CT being the clinical standard for surgical planning) and an *anomaly gap* (near-complete absence of Lumbosacral Transitional Vertebra (LSTV) annotation in existing CT datasets despite a 5–35% population prevalence). We introduce CTSpinoPelvic1K, a CT-native benchmark of 1,137 colonoscopy-protocol abdominal CT volumes across 804 patients with a unified 10-class spinopelvic segmentation labelmap and dual-provenance LSTV phenotype labels. The benchmark is assembled from CTSpine1K and CTPelvic1K—two independently curated public datasets sharing a patient cohort but using incompatible NIfTI coordinate conventions—via a two-phase patient-anchored pipeline: canonical UID matching resolves identity with algebraic certainty, after which intrapatient exhaustive bone-HU maximization selects the source CT series for each annotation with no heuristic fallbacks. A key methodological contribution is a *partial-annotation training protocol*: only 338 of the 1,137 cases carry fused spine-plus-pelvis ground truth, but we describe how the remaining 799 spine-only and pelvic-only cases can be incorporated into training via nnU-Net v2's `ignore_label` mechanism by rewriting unobserved background voxels, yielding a 3.36× increase in effective training data without biasing the model toward predicting background on unobserved structures. We further characterize systematic 0% cross-source agreement between two independent radiology teams at the sacralization boundary—not annotation noise but radiologically meaningful definitional indeterminacy between morphological and vertebral-counting operationalizations—and release both annotations with explicit arbitration metadata. A zero-shot TotalSegmentator benchmark concretely quantifies the anomaly gap: junction Dice drops 19.6 points (0.770 → 0.574) from Normal to Any-LSTV cases, with lumbarization cases degrading to 0.421 junction Dice owing to TotalSegmentator's structural inability to emit an L6 class. The CTSpinoPelvic1K dataset is publicly released on HuggingFace; the preprocessing pipeline, conversion scripts, and nnU-Net training infrastructure will be released under a permissive license upon acceptance.

Keywords: Spine Segmentation Benchmark, Partial-Annotation Training, Lumbosacral Transitional Vertebra, Multi-Dataset Label Fusion, Annotation Arbitration, TCIA, Surgical Planning, CT

Data and Code Availability The CTSpinoPelvic1K dataset (images, unified masks, and dual-provenance LSTV metadata) is derived from public TCIA collections under their respective data use agreements and is publicly released at https://huggingface.co/datasets/anonymous-mlhc/CTSpinoPelvic1K. The preprocessing pipeline, conversion scripts, and nnU-Net training infrastructure are not currently public; they will be released under a permissive open-source license upon acceptance.

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IRB All scans from de-identified public TCIA repositories.

1. Introduction

The lumbosacral junction is among the most clinically consequential regions of the axial skeleton. Surgical procedures at this level—transforaminal lumbar interbody fusion (TLIF), pedicle screw instrumentation, laminectomy, and sacropelvic fixation—require preoperative planning at millimeter accuracy, and errors at the lumbosacral level carry the highest medicolegal burden in spine surgery (Carrino et al., 2011). Accurate automated segmentation of the lumbar vertebrae, sacrum, and pelvic girdle supports pedicle screw trajectory optimization, sacropelvic parameter measurement, intervertebral implant sizing, and intraoperative neuronavigation registration.

The LSTV problem. A Lumbosacral Transitional Vertebra (LSTV) is a vertebra at the L5–S1 junction exhibiting morphological characteristics of the adjacent segment (**?**). *Sacralization* occurs when the fifth lumbar vertebra undergoes bony fusion with the sacrum, reducing the mobile lumbar spine to four segments. *Lumbarization* occurs when the first sacral segment detaches, mimicking an independent sixth lumbar vertebra. Prevalence estimates range from 5% to 35% depending on imaging modality and classification criterion (Sekharappa et al., 2014). Wrong-level surgery attributable to transitional anatomy misidentification occurs in an estimated 4.5–15% of cases where preoperative level counting relied on plain film or MRI alone (Konin and Walz, 2010).

The modality gap. The most widely used spine segmentation benchmarks—SPIDER, Spine Generic, and TotalSpineSeg—are built entirely on MRI (Graaf et al.; Cohen-Adad et al., 2021; Warszawer et al., 2025). Castellvi morphology is defined by cortical bridging and transverse process ossification, features CT resolves at sub-millimeter resolution and MRI does not. A model trained on MRI and deployed on preoperative CT acquires an implicit domain shift that degrades lumbosacral boundary delineation at the junction where clinical decisions are made.

The anomaly gap. CT-trained benchmarks including CTSpine1K (Deng et al., 2021), VerSe (Sekuboyina et al., 2020), TotalSegmentator-CT (Wasserthal et al., 2023), and CT-Pelvic1K (Liu et al., 2021) do not report or control for LSTV prevalence, do not stratify splits by transitional anatomy, and make no attempt to evaluate per-type performance at the lumbosacral junction. No prior ML benchmark has curated LSTV in CT with dense spinopelvic segmentation labels at scale, nor analyzed disagreement structure between independent LSTV annotation operationalizations.

The annotation uncertainty problem. Even when LSTV labels are attempted, independent radiologists applying different operationalizations of the same morphological concept can systematically disagree at the lumbosacral boundary. Existing datasets resolve this by publishing a single adjudicated label, discarding the disagreement signal. The medical machine learning community increasingly recognizes that inter-rater disagreement is signal, not noise (Dawid and Skene, 1979): CTSpinoPelvic1K applies this principle specifically to the LSTV confusion zone, where disagreement reflects genuine definitional indeterminacy rather than annotator error.

The partial-annotation opportunity. A systematic side-effect of fusing independently curated annotation sets is that most cases are *partially* annotated: the spine mask is present without a matching pelvic mask, or vice versa. Prior work typically discards these, retaining only the intersection where both annotations exist. For CTSpinoPelvic1K, this would collapse a 1,137-case benchmark to just 338 usable training cases—a two-thirds reduction that especially penalizes rare subgroups. We describe how nnU-Net v2's native `ignore_label` mechanism enables principled training across all 1,137 cases by rewriting unobserved background voxels to 255, eliciting zero loss contribution from unobserved structures without biasing the model.

Contributions.

1. **CTSpinoPelvic1K dataset.** 1,137 abdominal CT volumes across 804 matched patients with unified 10-class spinopelvic masks and 53 LSTV-labeled cases spanning four phenotypes, released as a public HuggingFace dataset.

2. **Registration-free fusion of CTSpine1K and CTPelvic1K.** Canonical UID matching resolves patient identity algebraically (825/825 patients, 100%); intrapatient bone-HU maximization assigns each mask to its source CT series without heuristic metadata fallbacks or spatial registration. We characterize and correct a previously undocumented Y-axis coordinate-convention mismatch that would silently corrupt all fused masks.

3. **Partial-annotation training protocol.** We describe an nnU-Net v2 `ignore_label` protocol that enables a 3.36× effective training-set expansion on this multi-source setting by using all 1,137 cases rather than only the 338 fully fused ones. Trained-baseline evaluation is deferred to future work (Section 9).

4. **Dual-provenance LSTV labels with arbitration metadata.** Retention of both CTPelvic1K morphological and CTSpine1K counting-based LSTV labels, with explicit provenance flags. We document 0% cross-source sacralization agreement— not annotation noise but definitional indeterminacy between two legitimate operationalizations— making the confusion zone explicit and available for uncertainty-aware training.

5. **Zero-shot TotalSegmentator benchmark quantifying the anomaly gap.** The most widely deployed multi-organ CT segmentation model in production use loses 19.6 junction-DSC points from Normal to Any-LSTV cases (0.770 → 0.574), collapsing to 0.421 on lumbarization cases owing to the absence of an L6 output class.

Generalizable Insights about Machine Learning in the Context of Healthcare

(1) Partial-annotation training via ignore-label materially expands the usable training set when fusing federated annotation sources. On CTSpinoPelvic1K, restricting training to fused cases would discard 70% of the annotated data; ignore-label training uses all 1,137 cases with principled supervision masking. The pattern applies directly to any multi-source dataset where different teams annotate different structures on the same underlying images.

(2) Inter-rater disagreement at anatomical boundaries is a recoverable signal, not noise. Two independent radiology teams applying legitimate but distinct operationalizations of the same morphological concept achieve 0% cross-source agreement on sacralization. Datasets should retain both annotations with provenance flags; downstream training and evaluation can then exploit rather than discard this disagreement.

(3) Coordinate-convention validation is a prerequisite for multi-source medical image fusion. The Y-axis reflection between CTSpine1K and CTPelvic1K is undetectable from voxel spacing alone and would have silently corrupted all fused masks. Any pipeline combining independently processed NIfTI datasets must explicitly validate affine-derived world coordinates on representative landmarks before training.

2. Related Work

CT spine and pelvis benchmarks. CTSpine1K (Deng et al., 2021) annotated 1,005 multi-source CTs with per-vertebra segmentation, and VerSe (Sekuboyina et al., 2020) contributed 460 scans in challenge format. Neither reports LSTV prevalence or stratifies evaluation by transitional anatomy, and many VerSe-derived scans are cropped to a narrow spinal field of view, discarding the pelvic anatomy required for spinopelvic parameter computation. TotalSegmentator-CT (Wasserthal et al., 2023) labels 104 structures across 1,204 CTs, but provides no LSTV classification and reports neighboring-class confusion as a known failure mode at the lumbosacral junction. CTPelvic1K (Liu et al., 2021) encodes LSTV qualitatively in COLONOG mask filenames, which we leverage as a primary label source.

MRI benchmarks and the modality gap. SPIDER (Graaf et al.) provides 447 lumbar MRIs. TotalSpineSeg (Warszawer et al., 2025) is MRI-native; sacrum labels were bootstrapped via CT-to-MRI rigid registration, introducing residual domain noise at the lumbosacral boundary.

Multi-dataset label fusion and partial-annotation training. Li et al. (2025) demonstrated combining CTSpine1K and CTPelvic1K improves joint segmentation using a 3D residual encoder-decoder with linear self-attention at the bottleneck and cross-scale feature fusion in the decoder. That work treats the two datasets as independent sources, provides no LSTV-stratified evaluation, and does not resolve their incompatible coordinate conventions. On the methodological side, Wasserthal et al. (2023) and recent nnU-Net work (Isensee et al., 2024) have established that ignore-label training enables principled use of partial-annotation cases at scale. We are the first to describe this mechanism applied to the spine+pelvis multi-source fusion setting with explicit LSTV-stratified evaluation.

LSTV in the ML literature. Kwak et al. (2025) evaluated a CNN on lumbar radiographs for binary LSTV detection. BUU-LSPINE (Klinwichit et al., 2023) contributes 3,600 lumbar X-rays with LSTV annotation, demonstrating clinical demand for automated LSTV recognition that CTSpinoPelvic1K extends into 3D surgical planning CT. No prior ML benchmark has curated LSTV in CT with dense spinopelvic segmentation labels at scale.

Orthopedic trauma and oncological benchmarks. The PENGWIN challenge (Sang et al., 2025) and Spine-Mets-CT-SEG (Pieper et al., 2024) establish that per-pathology

Table 1: Comparison of spine, pelvis, and adjacent CT benchmarks. ✓ = available, — = not available, † = via cross-modal registration, ‡ = qualitative filename encoding only. CTSpinoPelvic1K is the only benchmark combining fused labels, LSTV annotation, dual-provenance arbitration, and a registration-free fusion pipeline applicable to arbitrary TCIA collection pairs.

Dataset	Mod.	N	Vertebral	Pelvic	Fused	LSTV	Dual-Prov.	Pathological Focus	CT-native
VerSe (Sekuboyina et al., 2020)	CT	460	✓	—	—	—	—	Normal anatomy	✓
CTSpine1K (Deng et al., 2021)	CT	1,005	✓	—	—	—	—	Normal anatomy	✓
CTPelvic1K (Liu et al., 2021)	CT	1,184	partial	✓	—	‡	—	Normal + metal artifacts	✓
TotalSegmentator (Wasserthal et al., 2023)	CT	1,204	✓	✓	✓	—	—	General multi-organ	✓
SPIDER (Graaf et al.)	MRI	447	✓	—	—	—	—	Normal anatomy	—
TotalSpineSeg (Warszawer et al., 2025)	MRI	1,404	✓	†	†	—	—	Normal anatomy	—
PENGWIN (Sang et al., 2025)	CT	150	—	✓	—	—	—	Pelvic fractures	✓
Spine-Mets-CT-SEG (Pieper et al., 2024)	CT	55	✓	—	—	—	—	Spinal metastases	✓
BUU-LSPINE (Klinwichit et al., 2023)	X-Ray	3,600	✓	—	—	✓	—	LSTV	—
CTSpinoPelvic1K (ours)	CT	**1,137**	✓	✓	✓	✓	✓	**LSTV / congenital variants**	✓

evaluation is required: population-average metrics mask failure modes on the clinically highest-risk cases.

3. Dataset Construction: Patient-Anchored Fusion of CTSpine1K and CTPelvic1K

CTSpine1K and CTPelvic1K share a common COLONOG patient cohort but were independently curated: masks are released as NIfTI files without the `SeriesInstanceUID` linkage to the primary DICOM data, and the two teams used different `dcm2niix` invocations, producing silently incompatible coordinate conventions (Section 5.3). Fusing these annotation sets therefore requires solving two problems: (i) *identity*—determining which pelvic mask belongs to which patient in which dataset—and (ii) *alignment*—determining which specific CT acquisition each mask was drawn from, in world coordinates. We solve both without spatial registration or heuristic metadata fallbacks.

Identity via canonical UID join. Both CTSpine1K and CTPelvic1K mask filenames encode the TCIA `PatientID` with zero-padded UID suffixes whose padding conventions differ between dataset releases, causing naive string matching to fail. Canonical UID matching (Eq. 1) reduces the problem to an $O(N)$ exact relational join, algebraically guaranteed correct up to padding. On CTSpine1K ∪ CTPelvic1K this resolves 825/825 patients (100%) with zero heuristic fallbacks.

Alignment via intrapatient bone-HU maximization. Restricting candidate series to the patient's own TCIA acquisitions reduces the search space from a global index ($\sim 10^5$–10^6 CT series) to $|\mathcal{S}_p| \approx 4$ candidates. Within this bounded set, bone-HU maximization (Eq. 2) serves as a metadata-free oracle for mask-to-series assignment—useful because the DICOM metadata fields normally relied upon (`SeriesDescription`, `BodyPartExamined`) are frequently missing or mislabeled in TCIA releases. Placement succeeds on 789/792 spine masks (99.6%) and 714/714 pelvic masks (100%); the failure modes are documented in Section 7.

Partial-annotation opportunity. Because CTSpine1K and CTPelvic1K annotate overlapping but non-identical patient subsets, the fused benchmark is *partially annotated*: 338 of 1,137 cases carry both spine and pelvic ground truth, 443 are spine-only, and 356 are pelvic-only. Prior work typically discards partial cases, retaining only the intersection. We instead describe an nnU-Net v2 `ignore_label` protocol for training across all 1,137 cases (Section 5.5), yielding a 3.36× effective training-set expansion relative to fused-only baselines.

Scope of generalization. The patient-anchored pipeline is specific to the CTSpine1K/CTPelvic1K fusion demonstrated here. The underlying primitives—UID canonicalization for identity resolution, and tissue-HU-based intrapatient series selection for alignment—are applicable whenever two annotation sets share a TCIA patient cohort, but we do not claim empirical validation beyond the single dataset pair reported here. We discuss candidate extensions in Section 9.

4. Source Data

CTSpinoPelvic1K assembles three interlocking public TCIA resources sharing a common patient cohort.

COLONOG. The CT Colonography collection (Smith et al., 2015) provides paired supine and prone abdominal CTs from 825 subjects. Protocol: 100–120 kVp, in-plane resolution 0.625–0.977 mm, slice thickness 0.8–1.0 mm, soft-tissue kernel. The paired acquisitions—absent from all prior spine benchmarks— enable position-invariant training.

CTSpine1K. CTSpine1K (Deng et al., 2021) annotated 784 COLONOG cases with per-vertebra segmentation using the VerSe label scheme (IDs 20–25 encode L1–L6; ID 26 encodes sacrum).

CTPelvic1K. CTPelvic1K (Liu et al., 2021) provides four-class pelvic masks (sacrum, left hip, right hip) for 714 of the same COLONOG patients. Despite originating from the same acquisitions, different `dcm2niix` invocations produced a systematic Y-axis reflection between label spaces (Section 5). CTPelvic1K mask filenames encode patient UID, SeriesNumber, and LSTV phenotype as a qualifier, providing the primary LSTV label source for CTSpinoPelvic1K.

5. Methods

5.1. Phase 1: Patient-Anchored Identity Resolution

Both CTSpine1K and CTPelvic1K mask filenames encode the TCIA `PatientID` DICOM tag but not the `SeriesInstanceUID`. Both UIDs carry zero-padded last components that vary between dataset releases, causing naive string matching to fail. We establish patient identity via canonical UID matching:

$$\text{canon}(u) = \text{prefix}(u) \,\|\, \text{int}(\text{suffix}(u)) \tag{1}$$

Table 2: COLONOG scan statistics for CTSpinoPelvic1K.

Property	Value
Patients total	825
With both positions	694
Spine masks (CTSpine1K)	784
Pelvic masks (CTPelvic1K)	714
Matched patients (CTSpinoPelvic1K)	804
Matched volumes (CTSpinoPelvic1K)	1,137
Fused (spine + pelvis)	338
Spine-only	443
Pelvic-only	356

where int$(\cdot)$ strips leading zeros from the final UID component. Identity is *algebraically exact*—no heuristic required. PatientDB (schema in Appendix A) organizes all data using canonical patient UID as the primary key, and resolves **825/825 patients (100%)**.

5.2. Phase 2: Intrapatient Exhaustive Series Assignment

For mask m belonging to patient p, we identify its source TCIA series by exhaustive intrapatient search. COLONOG patients have $|\mathcal{S}_p| \approx 4$ CT-quality series, making this tractable. For each candidate series, we compute bone-HU coverage:

$$f_{\text{bone}}(m, s) = \frac{\sum_{v \in \mathcal{L}(m,s)} \mathbf{1}[I_s(v) > \tau_{\text{bone}}]}{|\mathcal{L}(m, s)|} \quad \tau_{\text{bone}} = 200 \text{ HU} \tag{2}$$

and select $s^*(m) = \arg\max_{s \in \mathcal{S}_p} f_{\text{bone}}(m, s)$. The same criterion applies to both spine and pelvic masks, differing only in the geometric alignment function $\mathcal{L}(m, s)$. This eliminates metadata-dependent heuristics: a correctly matched spine mask exhibits $f_{\text{bone}} \gg 0.5$; placement over a soft-tissue or scout series yields $f_{\text{bone}} \approx 0$ (Figure 3). Algorithm 1 in Appendix C provides pseudocode.

5.3. Affine Placement: The Y-Axis Reflection

The DICOM standard mandates an LPS+ patient basis while NIfTI natively encodes RAS+. When independent teams convert the same DICOM source using different toolchains, the resulting NIfTI affines carry silently inverted axes. CTSpine1K encodes center voxels at world $y \approx -181$ mm; CTPelvic1K at $y \approx +181$ mm for the same patients. Given NIfTI affine A and voxel extent N_y:

$$y_{\text{ctr}} = A_{1,3} + \tfrac{N_y}{2} \cdot A_{1,1}, \quad \text{flip} = \left(y_{\text{ctr}}^{\text{mask}} \cdot y_{\text{ctr}}^{\text{ref}} < 0\right) \tag{3}$$

When flip = true, the mask array is reversed along axis 1 and the affine updated exactly. Detection succeeds on all 714 pelvic masks; zero failures. CTPelvic1K masks are additionally z-cropped sub-volumes; the starting z-offset is recovered by cross-correlating CT bone-density and mask occupancy profiles.

5.4. Unified Label Map

CTSpine1K VerSe labels and CTPelvic1K labels are remapped into a unified 10-class scheme: 0 = background; 1–5 = L1–L5 (VerSe 20–24); 6 = L6 / lumbarized S1 (VerSe 25; LSTV only); 7 = sacrum (VerSe 26 / CTPelvic1K label 1); 8 = left hip (CTPelvic1K 2); 9 = right hip (CTPelvic1K 3). Class 6 is the first evaluation class in any public benchmark reserved for the lumbarized transitional body. At voxel-level conflicts at the L5/sacrum boundary, CTPelvic1K takes precedence for sacral extent.

5.5. Partial-Annotation Training Protocol

Fusion status is derived post-hoc: a case is FUSED when $s^*(\text{spine}) = s^*(\text{pelvic})$, SPINE-ONLY when only a spine mask exists, and PELVIC-ONLY when only a pelvic mask exists. Training on fused cases alone would discard 799 of 1,137 volumes (70%), with a disproportionate penalty on rare LSTV subgroups. We instead define a training protocol incorporating all 1,137 cases via nnU-Net v2's `ignore_label` mechanism.

Concretely, for each case we rewrite the saved label volume as follows:

Config	Label rewrite
FUSED	identity (all 10 classes valid)
SPINE-ONLY	voxels= $0 \rightarrow 255$; classes 1–6 retained
PELVIC-ONLY	voxels= $0 \rightarrow 255$; classes 7–9 retained

The `ignore` class 255 is declared in the nnU-Net `dataset.json`; voxels at this value contribute zero to both the cross-entropy and the Dice loss, so the model is supervised only on structures present in each case's ground truth. For a SPINE-ONLY case, the model learns "here are L1–L6" without being penalized on the pelvic region; symmetrically for PELVIC-ONLY. Spine classes receive supervision from all 338 fused plus 443 spine-only cases (781 total); pelvic classes from 338 fused plus 356 pelvic-only cases (694 total). This is a principled use of federated annotation sources. The test set is restricted to fused cases so that per-class DSC, HD95, and mIOU are all computable on complete ground truth. Empirical evaluation of a trained nnU-Net baseline under this protocol is deferred to future work (Section 9).

6. LSTV Labels and the Lumbosacral Confusion Zone

6.1. Dual-Provenance Label Sources

CTPelvic1K filename qualifiers encode a *morphological criterion*: the annotator inspects for cortical bridging, transverse process hypertrophy, and pseudoarticulation with the sacral ala, assigning one of four classes (NORMAL, LUMBARIZATION, SEMI-SACRALIZATION, SACRALIZATION) based on Castellvi morphology. This is the canonical radiological definition.

CTSpine1K vertebral heuristics apply a *vertebral counting criterion*: VerSe label 25 present $\Rightarrow$ lumbarization; VerSe label 24 absent $\Rightarrow$ sacralization; labels 20–24 present without 25 $\Rightarrow$ normal.

Figure 1 illustrates representative cases for each LSTV phenotype.

Table 3: CTSpinoPelvic1K stratified splits. Training uses all configurations via ignore-label; validation and test restricted to fused cases for clean per-class evaluation.

Configuration	Train	Val	Test	Total
Fused (10 classes)	234	52	52	338
Spine-only (ignore)	309	72	68	449
Pelvic-only (ignore)	268	52	56	376
Total training volumes	**811**	**176**	**176**	**1,163**
LSTV-positive (any config)	36	9	9	54
LSTV-positive (fused only)	3	2	2	7

Table 4: Cross-source LSTV agreement. The 0% sacralization agreement quantifies the lumbosacral confusion zone.

LSTV Class	N	%	**Cross-val**	**Agreement**
Normal	1,084	95.3	100%	✓
Lumbarization	22	1.9	n/a[†]	—
Semi-sacralization	3	0.3	67%[‡]	partial
Sacralization	28	2.5	0%[§]	**0%**
Total LSTV	53	4.7	—	—
Total	1,137	100	—	—

[†] Pelvic protocol omits lumbarization body.
[‡] $n = 3$; 1 case UNKNOWN. [§] Definition mismatch; see text.

6.2. The Confusion Zone

Table 4 shows cross-source agreement. Sacralization shows **0% cross-source agreement** ($n = 28$ CTPelvic1K sacralization cases, zero overlap with CTSpine1K sacralization cases). This is not annotation error: a patient exhibiting Castellvi Type II morphology (bilateral pseudoarticulation satisfying the morphological criterion) can simultaneously retain five independently countable lumbar vertebral bodies if the bridging is partial. Both annotators are correct within their own criterion; the case sits precisely at the boundary of both classification systems (panel B of Figure 1).

Figure 2 illustrates the three anatomical configurations that produce systematic disagreement: Type A (partial bridging with retained L5 body; Castellvi Type II), Type B (asymmetric ossification), and Type C (definitional indeterminacy between "sacralized L5" and "lumbarized S1" without MRI correlation).

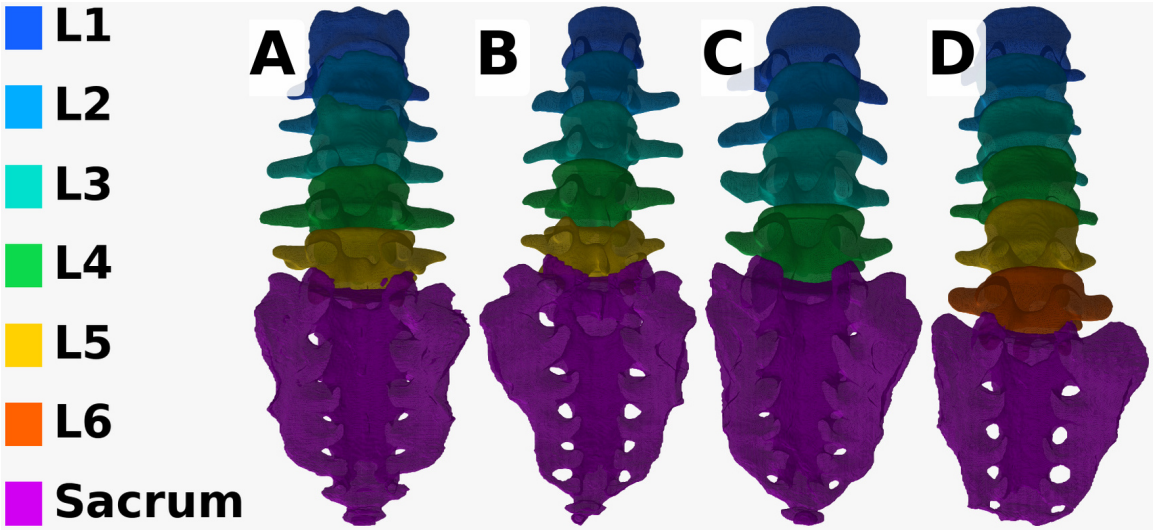


Figure 1: **Representative CTSpinoPelvic1K lumbosacral segmentation masks for each LSTV phenotype.** All renders are from fused cases displayed in anterior clinical view at native voxel resolution. Label colors follow the 10-class scheme: L1 (royal blue), L2 (sky blue), L3 (cyan-teal), L4 (green), L5 (yellow), L6 (orange), spine sacrum (red), pelvic sacrum (magenta). **(A)** *Normal anatomy* (COLONOG token 2): five lumbar vertebrae above the sacrum in standard configuration. **(B)** *Sacralization — morphological criterion* (COLONOG token 32): Castellvi-type sacral fusion encoded in the CTPelvic1K filename qualifier; the CTSpine1K vertebral counting criterion independently labels this case Normal, exemplifying the 0% cross-source sacralization agreement. **(C)** *Sacralization — vertebral-count criterion* (COLONOG token 107): VerSe label 24 absent; only four lumbar vertebrae identifiable. No CTPelvic1K morphological qualifier encodes this case as abnormal, again illustrating the definitional non-overlap. **(D)** *Lumbarization* (COLONOG token 167): VerSe label 25 (L6, orange) present as a discrete additional lumbar body between L5 (yellow) and the sacrum.

7. Dataset Quality Assessment

Bone fraction f_{bone} (Eq. 2) provides an automated series-conflict detector. Figure 3 shows the distribution across all 1,503 placed masks. Spine: mean 68.3%, SD 11.2%, min 15.2%. Pelvic: mean 66.2%, SD 8.9%, min 36.8%. 789/792 spine masks placed (99.6%); the 3 failures correspond to CTSpine1K annotations where no intrapatient series exceeded the bone-HU threshold, and these cases are excluded from spine-containing configurations only. All 714/714 pelvic masks placed (100%). Zero spine masks had fewer than four identifiable lumbar labels; 9 had exactly four (all concordant with CTPelvic1K sacralization or semi-sacralization qualifiers).

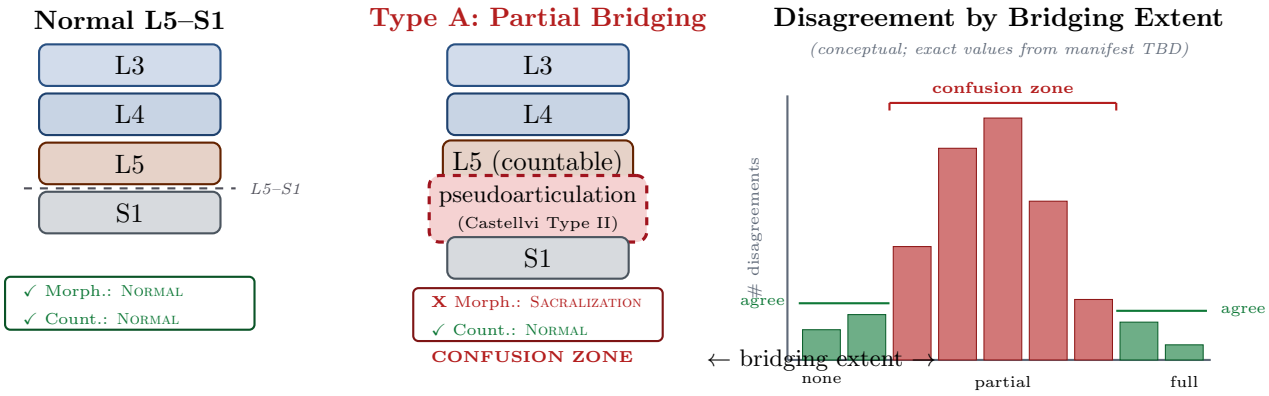


Figure 2: **The lumbosacral confusion zone. Left**: normal L5–S1 anatomy with discrete junction. **Center**: the Type A configuration that accounts for the majority of the 0% sacralization agreement: Castellvi Type II pseudoarticulation satisfies the morphological criterion (sacralization) while a discrete L5 body satisfies the counting criterion (normal). The transition zone (orange, dashed outline) is the voxel region at L5/S1 boundary whose class assignment is ambiguous. **Right**: conceptual disagreement rate as a function of cortical bridging extent. The confusion zone (red bars) is concentrated at intermediate bridging; full fusion and absent bridging produce reliable agreement. Exact values will be reported from the placed manifest.

8. Zero-Shot Benchmark: TotalSegmentator

We evaluate the most widely deployed multi-organ CT segmentation model in production clinical use as a zero-shot baseline on CTSpinoPelvic1K. This isolates the *anomaly gap*: the performance cost, at the lumbosacral junction, of a model whose training distribution and label vocabulary were not designed with LSTV in mind.

8.1. Inference Setup

Off-the-shelf inference with the pretrained `total` task (Wasserthal et al., 2023), executed in *fast mode* (`--fast`), which resamples the input volume to 3 mm isotropic resolution before prediction. Inference was run on a single NVIDIA L40S GPU (46 GB) across the full CTSpinoPelvic1K matched cohort ($N = 921$ volumes completing inference: 272 fused, 340 spine-only, 309 pelvic-native), of which 46 are LSTV-positive (26 sacralization, 16 lumbarization, 4 other). We remap TotalSegmentator's native output labels (`vertebrae_L1`–`vertebrae_L5`, `sacrum`, `hip_left`, `hip_right`) to our 10-class scheme. TotalSegmentator does not emit an L6 class, so class 6 performance is zero by construction; for lumbarization cases whose ground truth contains an L6 body, this contributes a hard failure mode that directly quantifies the clinical gap. Because TotalSegmentator was not trained on any CTSpinoPelvic1K case, we evaluate it zero-shot on the full matched cohort rather than

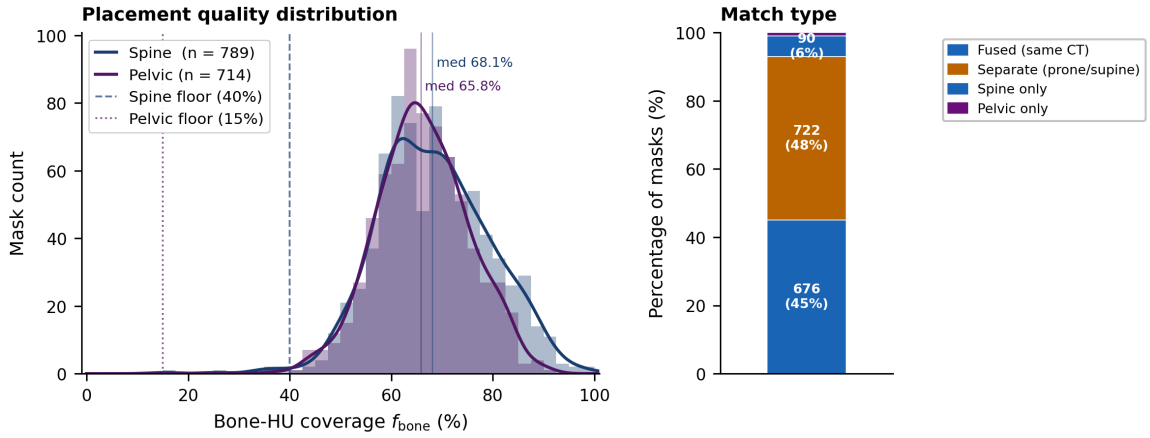


Figure 3: **Bone-HU coverage distribution across 1,503 placed masks.** Spine ($N = 789$, blue) and pelvic ($N = 714$, orange) both show strong bimodal separation between correctly matched bone-protocol CT series ($f_{\text{bone}} > 0.5$; dominant right mode) and mismatched or scout-image series ($f_{\text{bone}} \approx 0$; small left mode). The gap between modes validates bone-HU maximization as a series-selection criterion without any metadata dependence.

Table 5: TotalSegmentator zero-shot per-class DSC (%) on the full CTSpinoPelvic1K fused cohort ($n = 272$), fast mode (3 mm). Junction DSC is evaluated over a 40 mm axial window centered on the L5/S1 boundary. Class 6 (L6) is zero by construction: TotalSegmentator does not emit an L6 class.

Model	L1–L5	L6	Sacrum	Jxn DSC
TotalSegmentator	85.4	0.0*	77.4	76.6

* TotalSegmentator does not emit an L6 class (structurally zero).
L1–L5 is the mean of L1=84.7, L2=84.6, L3=85.7, L4=86.0, L5=85.9.

restricting to the 52-case test split; this materially increases the statistical power of the LSTV-subgroup analysis (Section 8.3). The 3 mm fast-mode setting trades approximately 1–2 DSC points of headline performance (Wasserthal et al., 2023) for ∼20× wall-clock speedup; we report fast-mode numbers as a realistic deployment proxy, noting that the LSTV performance gap reported below is a property of TS's label vocabulary (no L6 class) and training distribution, not of the resampling resolution.

8.2. Per-Class Performance on the Full Fused Cohort

Table 5 reports per-class DSC on the full fused cohort ($n = 272$).

Table 6: TotalSegmentator zero-shot performance stratified by LSTV phenotype (fast mode, 3 mm). Jxn DSC = 40 mm axial-window Dice at the L5/S1 boundary; ErrRate = per-voxel class-mismatch fraction within that window; TS_L6 = count of cases in which TS emitted any L6 voxel (structurally zero by label vocabulary). The 19.6-point junction-DSC gap between Normal and Any-LSTV concretely quantifies the clinical cost of the anomaly gap.

Subgroup	n	**Jxn DSC**	**ErrRate**	**TS_L6**
Normal	875	0.770	0.330	0
Any LSTV	46	0.574	0.317	0
Sacralization	26	0.651	0.213	0
Lumbarization	16	0.421	0.526	0

TotalSegmentator achieves 85.4% mean L1–L5 Dice and 77.4% sacrum Dice on the full fused cohort, but drops to 76.6% at the L5/S1 junction window—and, critically, collapses further on LSTV-positive cases (Section 8.3). This baseline is well motivated: it is the most widely deployed multi-organ CT segmentation model in production clinical use, and its failure mode on CTSpinoPelvic1K's LSTV subgroup constitutes the concrete anomaly gap our benchmark is designed to close.

8.3. LSTV Subgroup Analysis

Because TotalSegmentator is zero-shot on CTSpinoPelvic1K, every matched case is a valid test case; we therefore evaluate its LSTV-subgroup performance on the full matched cohort rather than the 52-case fused test split. This yields 46 LSTV-positive cases (26 sacralization, 16 lumbarization, 4 other) alongside 875 normal-anatomy cases—a sample size sufficient to resolve phenotype-stratified performance differences at the junction. Table 6 reports the stratified analysis.

Three findings stand out. First, the junction DSC gap between Normal (0.770) and Any-LSTV (0.574) is 19.6 points—a clinically large disparity at the exact anatomic level where wrong-level surgery risk is concentrated. Second, the disparity is driven asymmetrically by phenotype: sacralization cases retain 0.651 junction DSC, whereas lumbarization cases collapse to 0.421. The gap is mechanistic rather than mysterious: every lumbarization case carries a ground-truth L6 body that TS cannot emit (`TS_L6`=0 across all 16 cases), so the L5/S1 window is guaranteed to contain a voxel-count proportion of unrecoverable error. Third, per-vertebra Dice degrades monotonically from L1 (58.6%) to L5 (37.2%) within the Any-LSTV subgroup, confirming that the loss of performance concentrates at the caudal lumbar levels rather than being distributed uniformly across the spine.

Per-case qualitative predictions for the 9 LSTV-positive cases in the fused test split, together with agreement-status stratification, are provided in Appendix D.

9. Discussion

Partial-annotation training unlocks the full data scale. For federated annotation sources—the normal case when fusing independently curated datasets—discarding partial cases imposes a silent two-thirds data penalty at the CTSpinoPelvic1K scale. Ignore-label training lets every annotated voxel contribute supervision only where its class is observed. The pattern generalizes beyond spine/pelvis fusion: any multi-source dataset where different teams annotate different structures on the same underlying images can apply this mechanism directly.

The confusion zone is a scientific contribution. The 0% cross-source sacralization agreement documents, at scale, that two legitimate expert-level annotation strategies for the same anatomical feature produce non-overlapping classifications at the Castellvi Type I–II boundary. No prior spine ML benchmark has reported this because no prior benchmark retained both annotation sources with provenance. The `lstv_agreement` flag makes the ambiguity explicit, enabling uncertainty-aware training and calibration benchmarking on known-ambiguous cases.

The anomaly gap is empirically measurable. The zero-shot TotalSegmentator benchmark directly quantifies the clinical cost of the anomaly gap: the most widely deployed multi-organ CT segmentation model loses 19.6 junction-DSC points from Normal to Any-LSTV cases, and 34.9 points on lumbarization specifically ($0.770 \rightarrow 0.421$). Of the lumbarization-case error, a mechanically identifiable fraction traces to a single architectural decision—the absence of an L6 output class—illustrating how label vocabulary choices made during benchmark design propagate as hard failure modes years later in deployment.

Implications for surgical planning AI. A segmentation model that outputs high uncertainty at L5/S1 for a Castellvi Type II case is more useful to a surgeon than one that outputs a hard incorrect label with high confidence. CTSpinoPelvic1K provides the annotation structure necessary to train and evaluate such models.

The coordinate convention problem is general. The Y-axis reflection between CTSpine1K and CTPelvic1K is an instance of a broad failure class: whenever independent teams convert the same DICOM source using different toolchains, silently inverted NIfTI axes result. Any multi-source fusion pipeline must validate affine-derived world coordinates for representative landmarks.

Limitations. CTSpinoPelvic1K derives from colonoscopy-protocol CT (soft-tissue kernel), which may not generalize to bone-kernel protocols used in some surgical planning workflows. LSTV labels are inherited without independent re-reading; the 0% sacralization agreement highlights sensitivity to annotation protocol. Semi-sacralization has limited representation ($n = 3$). TotalSegmentator was evaluated in fast mode (3 mm isotropic resampling); full-resolution (1.5 mm) inference would be expected to recover 1–2 DSC points on standard anatomy, but the label-vocabulary-driven lumbarization failure mode is resolution-invariant. The present submission does not include a trained nnU-Net baseline; the infrastructure for partial-annotation training is described in Section 5.5 and the training pipeline is fully implemented, but empirical evaluation is deferred to future work. Consequently, the head-to-head comparison of the ignore-label protocol against fused-only

training is described but not empirically validated. The COLONOG cohort (colonoscopy-screening age, 45–75, US academic centers) may not represent the full distribution of spine morphology in surgical planning populations.

Future directions. Immediate next steps include: training and evaluating the nnU-Net v2 baseline under the partial-annotation protocol described in Section 5.5; running a fused-only vs. all-cases ablation to isolate the contribution of ignore-label training; and benchmarking additional architectures—including the cross-scale attention architecture of Li et al. (2025) and other recent spine-pelvis fusion models—on the released benchmark. Within the spine domain, further extensions include Castellvi sub-type classification (Type I–IV), bone-kernel protocol inclusion, and validation on LSTV-enriched external cohorts with independent re-reading. The patient-anchored pipeline introduced here is also applicable in principle to other TCIA collection pairs that share a patient cohort; empirical validation on such extensions is beyond the scope of the present work.

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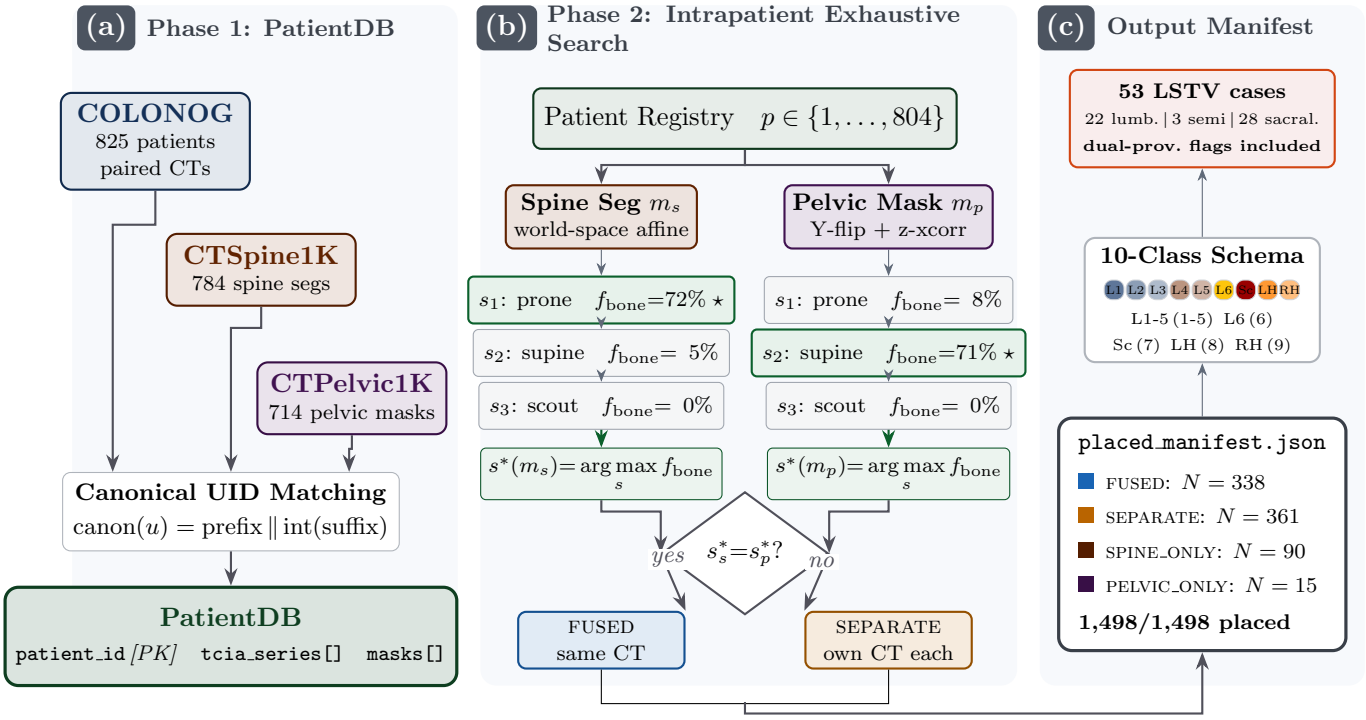


Figure 4: **Two-phase CTSpinoPelvic1K construction pipeline. (a)** Phase 1: canonical UID matching builds PatientDB. **(b)** Phase 2: bone-HU exhaustive search assigns each mask to its optimal series; outcomes merge into the manifest. **(c)** The manifest feeds the 10-class schema and dual-provenance LSTV summary. $\star$ = bone-HU winner.

Appendix A. PatientDB Schema

PatientDB organizes all data using canonical patient UID as the primary key, with five entities: `PATIENT` (root entity indexed by canonical UID); `TCIA_SERIES` (CT acquisitions per patient); `SPINE_MASK` and `PELVIC_MASK` (annotation entities from CTSpine1K and CTPelvic1K respectively); and `PLACED_CASE` (output rows in the final manifest, linking a mask to its selected series and fusion type). The separation of identity (Phase 1) from alignment (Phase 2) reduces a globally intractable matching problem to a locally bounded one per patient, and generalizes immediately to any TCIA collection pair sharing a patient cohort.

Appendix B. Pipeline Construction (Detailed)

Figure 4 illustrates the two-phase construction pipeline in full. All styles in the TikZ diagram use pure named colors as style arguments; no compound color (of the form `A!n`) is passed as a parameterized argument.

Appendix C. Algorithms and Implementation

Algorithm 1 Phase 2: Intrapatient Exhaustive Series Assignment

Require: PatientDB $\mathcal{D}$, alignment function $\textsc{Align}(m, s)$ returning placed voxels $\mathcal{L}$
Require: Bone-HU threshold $\tau = 200$ HU
Ensure: Placement manifest $\mathcal{M}$

for each patient $p \in \mathcal{D}$ **do**
  $\mathcal{S}_p \leftarrow$ CT-quality series for patient p
  for each mask $m \in \{\textsc{SpineMasks}(p) \cup \textsc{PelvicMasks}(p)\}$ **do**
    $f^* \leftarrow 0$, $s^* \leftarrow \varnothing$
    for each series $s \in \mathcal{S}_p$ **do**
      $\mathcal{L} \leftarrow \textsc{Align}(m, s)$
      $f \leftarrow |\{v \in \mathcal{L} : I_s(v) > \tau\}|/|\mathcal{L}|$
      if $f > f^*$ **then**
        $f^* \leftarrow f$, $s^* \leftarrow s$
      end if
    end for
    $\mathcal{M}[p][m] \leftarrow (s^*, f^*)$
    Persist aligned mask $\mathcal{L}^*$ to `placed/`
  end for
end for
return $\mathcal{M}$

NIfTI orientation convention. All placed masks are saved in PIR (Posterior-Inferior-Right) orientation, matching MONAI `Orientationd(axcodes='PIR')` at training time.

TotalSegmentator inference configuration. TotalSegmentator v2 (`total` task) was run in fast mode (`--fast`), which triggers the 3 mm isotropic-resampling model rather than the default 1.5 mm variant. Inference was executed per case on a single NVIDIA L40S GPU (46 GB); typical per-case wall-clock time was 20–60 seconds, dominated by the two-stage resample/predict loop. Of the 1,137 matched volumes, 921 completed inference successfully and are reported in Tables 5 and 6; failures were attributable to input-header edge cases and are excluded rather than imputed. Output labels were remapped one-to-one onto the 10-class CTSpinoPelvic1K scheme, with TS `sacrum` → class 7, `vertebrae_L1`–`L5` → classes 1–5, `hip_left`/`hip_right` → classes 8/9. Class 6 (L6) has no TS source class by construction.

Partial-annotation label rewrite. The conversion script `tools/convert_hf_to_nnunet.py` reads each case's label NIfTI; for SPINE-ONLY and PELVIC-ONLY cases, voxels with value 0 are rewritten to 255 (the declared `ignore_label`) before being written to `labelsTr/`. FUSED cases are copied verbatim. The resulting `dataset.json` declares: `"labels":{"background":0,...,"right_hi`

Planned nnU-Net configuration. The infrastructure for the future trained baseline uses nnU-Net v2 with the ResEncM planner, GPU memory target 100 GB, overwritten plans name `nnUNetResEncUNetPlans_100G`, configuration `3d_fullres`, and a 100-epoch trainer variant. Loss: `DC_and_CE_loss` (nnU-Net default). Precision: bf16 mixed. One H200 GPU

(140 GB HBM3e). Fold 0 only; the `splits_final.json` is generated by the conversion script to match our train/val/test assignment exactly.

Dual-provenance flag schema. Each manifest record includes: `lstv_pelvic` (CTPelvic1K morphological label); `lstv_vertebral` (CTSpine1K counting heuristic); `lstv_agreement` (Boolean: both sources agree on non-Normal classification); `lstv_confusion_zone` (Boolean: sources disagree on LSTV status, marking a confusion-zone case).

Appendix D. LSTV Case Details

The 9 LSTV-positive test cases are distributed as: 2 fused (1 lumbarization, 1 sacralization), 3 spine-only (1 lumbarization, 2 sacralization), and 4 pelvic-native (2 lumbarization, 2 sacralization). Per-case qualitative predictions, agreement status, and junction segmentation overlays are provided with the released dataset as `lstv_test_qualitative.pdf`; the small cohort size precludes reporting sensitivity/specificity with meaningful confidence intervals here.